

# Wang Jiefan

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## EDUCATION

### Bachelor of Computing

Honours w. Distinction (CGPA: 4.33) - Focus in Parallel Computing and Database Systems

National University of Singapore

Aug. 2021 – July 2025

## EXTRA-CURRICULAR

### Team Kent Ridge (High-Performance Cluster Computing Interest Group)

June 2024 - July 2025

- Co-founder, captain of competition group of 20+ students.
- Designed and built private multi-node research cluster, administered with Ansible, running on Rocky Linux, serving a Gluster-distributed XFS filesystem and integrated with the Spack package manager.
- Wrote whitepapers on HPC technologies such as CUDA GPGPU programming, advanced C++ features and memory-aware software optimization.

## ACHIEVEMENTS

### SC24 Indy-Student Cluster Competition

July - Nov. 2024

- Highest HPLinpack Score** (22.6 TFLOPS across 30 AMD EPYC-Milan CPU Nodes).
- Implemented MPI topology optimizations, kernel patches and runtime optimizations for the HPL Linear algebra benchmark and Nanoscale Molecular Dynamics simulation.

### ISC25 HPC-AI Student Cluster Competition

Aug. 2024 - June 2025

- 3rd Place** (Online Category).
- Wrote patches for the `μphys` module of ICON (Icosahedral Nonhydrostatic Weather and Climate Model) using C++26 `std::par` compiler assisted offloading ( $\times 257$  runtime improvement,  $>4\text{hrs} \rightarrow 56.1\text{s}$ ).

## PROFESSIONAL EXPERIENCE

### Backend Software Engineer - Global E-Commerce (Europe)

Singapore

TikTok Pte. Ltd

June 2025 - Present

- Implemented performance optimizations, such as asynchronous querying and task scheduling policies, to business logic of the TikTok shop seller platform that resulted in a  $>60\%$  YoY increase in quarterly onboarding rate across all EU regions.
- Service OnCalls to maintain the stability and security of relevant systems, including performing data center disaster recovery and loss prevention/control.

### Backend Platform Developer Intern

Singapore

Shopee Pte. Ltd.

May - Aug. 2024

- Developed Virtual Gateway feature for internal container networking platform in Go using gRPC and etcd, to automate and segregate the allocation of subnets to hundreds of in-production Virtual Private Clouds (VPCs) across global business units.
- Implemented Quality of Service (QoS) policies for internal container networking platform, by dynamically setting Differentiated Services Code Point values in IP headers using cgroup-attached eBPF programs, to enable client-side configuration of TCP traffic priorities for hundreds of baremetal VPC servers.

### Software Engineer Intern

Singapore

Pinvest Pte. Ltd.

May - July 2023

- Designed and implemented parallelization and optimization constructs, such as streaming, pipelining and caching, into LLM service infrastructure. End-to-end latency for API endpoints improved by as much as 35%.
- Triaged and fixed several request-spoofing security vulnerabilities to mobile financial trading platform.

### Teaching Assistant

Singapore

National University of Singapore

Aug. 2023 - May. 2025

- CS2109S Introduction to Machine Learning (AY23/24 Sem 1), CS3211 Parallel and Concurrent Programming (AY23/24, AY24/25 Sem 2), CS3210 Parallel Computing (AY24/25 Sem 1/2)
- Administered 30-node Slurm-based student compute cluster, backed by a Gluster replicated filesystem and Prometheus+Grafana monitoring.

## PROJECTS

### The Conq Programming Language | Rust, WebAssembly

Jan. – April 2024

- General-purpose programming language implemented with LLVM-MLIR toolchain and compiles to WASM with HTML Canvas bindings.

### Order-Matching System | C++

Sep. – Nov. 2023

- Socket-based HFT-inspired multi-threading [order-matching engine](#) which utilizes custom extension of `std::priority_queue` that is thread-safe, supports arbitrary removal and iteration.

### PySlidingWindow | Python

Aug. – Nov. 2022

- Experimental lost-tolerant automatic-repeat-request (ARQ) protocol for reliable file transfer over UDP.  $\sim 15\text{kbps}$  (19% slowdown) at 40% packet corruption and 40% packet loss.

... and many more, at [nafeij.me](https://nafeij.me) and [github.com/Nafeij](https://github.com/Nafeij)

## TECHNICAL SKILLS

**Languages:** C/C++, Java, Kotlin, Go, Python, Typescript, Rust, PostgreSQL, SQLite, HTML\CSS, WebAssembly

**Frameworks:** React, OpenMP, OpenACC, React Native, Node.js, Next.js, gRPC, Flask, Ansible, Spack, Django, JUnit, NumPy, Pandas, PyTorch, Cilium

**Technologies:** Git, OpenMPI, eBPF, etcd, Kubernetes, ROCm, CUDA, Nvidia HPC SDK, Slurm, PBS, Singularity, Gluster, GraphQL, Jenkins, Firebase, OpenStack, JIRA

## MISCELLANEOUS

I am a **Singaporean citizen**.